OS LAB 2

Write a C program to simulate the following CPU scheduling algorithms to find the average turnaround time and waiting time using ***FCFS***.

```
#include<stdio.h>
#include<stdlib.h>

#define max 10

struct FCFS
{
    int pid;
    int at;
    int bt;
    int ct;
    int tat;
    int wt;
};

void sortAT(struct FCFS p[], int n)
{
    struct FCFS temp;
    for(int i=0;i<n-1;i++)
    {
        for(int j=i+1;j<n;j++)
        {
            if(p[i].at > p[j].at)
            {
                temp = p[i];
                p[i] = p[j];
                p[j] = temp;
            }
        }
    }
}

void computeCT(struct FCFS p[], int n)
{
    int ct = 0;

    for(int i=0;i<n;i++)
    {
        if(ct < p[i].at)
```

```
            ct = p[i].at;

        ct += p[i].bt;
        p[i].ct = ct;
    }
}

float calculateTAT(struct FCFS p[], int n)
{
    int totaltat = 0;

    for(int i=0;i<n;i++)
    {
        p[i].tat = p[i].ct - p[i].at;
        totaltat += p[i].tat;
    }

    return (float)totaltat / n;
}

float calculateWT(struct FCFS p[],int n)
{
    int totalwt = 0;

    for(int i=0;i<n;i++)
    {
        p[i].wt = p[i].tat - p[i].bt;
        totalwt += p[i].wt;
    }

    return (float)totalwt / n;
}

void display(float tat, float wt, struct FCFS p[], int n)
{
    if(!n)
    {
        printf("\nNo processes");
        return;
    }

    printf("\nProcess Details:");
    printf("\nPID\tAT\tBT\tCT\tTAT\tWT");

    for(int i=0;i<n;i++)
```

```c
    {
        printf("\n%d\t%d\t%d\t%d\t%d\t%d",
        p[i].pid,p[i].at,p[i].bt,p[i].ct,p[i].tat,p[i].wt);
    }

    printf("\n\nAverage Turnaround Time: %.2f",tat);
    printf("\nAverage Waiting Time: %.2f\n",wt);
}

int main()
{
    struct FCFS p[max];
    int n;

    printf("Enter number of processes: ");
    scanf("%d",&n);

    for(int i=0;i<n;i++)
    {
        p[i].pid = i+1;

        printf("\nEnter Arrival Time for P%d: ",i+1);
        scanf("%d",&p[i].at);

        printf("Enter Burst Time for P%d: ",i+1);
        scanf("%d",&p[i].bt);
    }

    sortAT(p,n);

    computeCT(p,n);

    float tat = calculateTAT(p,n);
    float wt = calculateWT(p,n);

    display(tat,wt,p,n);

    return 0;
}
```

OUTPUT:

```
Enter number of processes: 5

Enter Arrival Time for P1: 0
Enter Burst Time for P1: 3

Enter Arrival Time for P2: 2
Enter Burst Time for P2: 2

Enter Arrival Time for P3: 3
Enter Burst Time for P3: 5

Enter Arrival Time for P4: 4
Enter Burst Time for P4: 4

Enter Arrival Time for P5: 6
Enter Burst Time for P5: 1

Process Details:
PID     AT      BT      CT      TAT     WT
1       0       3       3       3       0
2       2       2       5       3       1
3       3       5       10      7       2
4       4       4       14      10      6
5       6       1       15      9       8

Average Turnaround Time: 6.40
Average Waiting Time: 3.40
```